

The irrationality exponent of nonzero rational arctangents is 2

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This manuscript presents an extension of OpenAI’s weighted logarithmic interpolation and determinant method to arctangents of nonzero rational numbers. The arithmetic takes place in the Gaussian integers. The argument retains the actual exponential factors in the matrix, the row denominator savings, and the approximation-center translation costs.

The Lean endpoints prove the statement below without an irrationality, non-torsion, interpolation, ampleness, or approximation-bound hypothesis. The geometric comparison framework and its formal infrastructure are inherited from OpenAI’s released library and our preceding rational-logarithm implementation. Substantial inherited lemmas are identified explicitly below. This is a mathematical exposition of the formal argument, not a claim of external human peer review.

Yufei Bai’s preprint submitted on 20 August 2026 reports the upper bound 8.585166... for the irrationality measure of $\arctan(1/2)$. We cite it as a concrete earlier benchmark; this investigation does not establish an exhaustive priority claim. [Bai, arXiv:2608.19782](#).

1. Statement and conventions

Theorem. For every rational number $r \neq 0$, the real number $\arctan r$ is irrational and

$$\mu(\arctan r) = 2.$$

In particular, $\mu(\arctan(1/2)) = 2$. The approximation statement proved before taking the irrationality exponent is

$$\forall \nu > 2 \exists Q \in \mathbb{Z}, Q \geq 2 : \quad \forall p, q \in \mathbb{Z}, q \geq Q \implies q^{-\nu} \leq \left| \arctan r - \frac{p}{q} \right|. \quad (1)$$

The threshold depends on r, ν , and is uniform in p, q . Fractions in (1) need not be reduced. The case $r = 0$ is excluded: its arctangent is zero and exact rational approximations prevent (1).

For an irrational real number x , $\mu(x)$ is the supremum of the positive exponents ν for which infinitely many reduced rational numbers p/q , with $q \geq 2$, satisfy $0 < |x - p/q| < q^{-\nu}$. The formal proof establishes irrationality and the lower bound 2 before identifying the supremum, so an unbounded-supremum convention plays no role.

We first prove a slightly broader angle statement. Let $\xi \neq 0$ be real, let $U \in \mathbb{Z}[i]$, and let D be a positive integer. Suppose

$$\alpha = U/D = e^{i\xi}, \quad \alpha^0, \alpha^1, \alpha^2, \dots \text{ are pairwise distinct.} \quad (2)$$

Then (1), with ξ in place of $\arctan r$, holds. The arctangent hypotheses and the passage from $2\arctan r$ to $\arctan r$ are discharged in §§2 and 8.

2. The arctangent exponential and the torsion exceptions

For rational r , put

$$\xi = 2 \arctan r, \quad \omega = i\xi, \quad a_r = \frac{1 - r^2}{1 + r^2}, \quad b_r = \frac{2r}{1 + r^2}.$$

The double-angle formulas and $\tan(\arctan r) = r$ give

$$e^{i\xi} = a_r + ib_r = \frac{1 + ir}{1 - ir}, \quad a_r^2 + b_r^2 = 1. \quad (3)$$

There is no choice of a global logarithm branch in this identity. We specify the real angle $2 \arctan r$ directly. The local logarithm used later is only $\log(1 + t)$ on $|t| < 1$.

Every number $a_r + ib_r$ can be written U/D with Gaussian U and positive integer D . The family implementation uses

$$D = \text{den}(a_r) \text{den}(b_r), \quad U = \text{num}(a_r) \text{den}(b_r) + i \text{num}(b_r) \text{den}(a_r).$$

Lowest terms are unnecessary. For the first target, the separate formal path uses exactly

$$r = 1/2, \quad \xi = 2 \arctan(1/2), \quad U = 3 + 4i, \quad D = 5. \quad (4)$$

We need the exact root-of-unity exceptions. Suppose $z = a + ib$, with $a, b \in \mathbb{Q}$, $a^2 + b^2 = 1$, and $z^n = 1$ for a positive integer n . Both z and $\bar{z}$ are algebraic integers. Hence

$$2a = z + \bar{z}, \quad 2b = -i(z - \bar{z})$$

are rational algebraic integers and therefore ordinary integers. Write them as k, l . Then $k^2 + l^2 = 4$, so (k, l) is one of $(\pm 2, 0), (0, \pm 2)$. Thus $z \in \{1, -1, i, -i\}$.

For (3), $a_r = 1$ forces $r = 0$; $a_r = -1$ is impossible; and $a_r = 0$ forces $r = \pm 1$. Conversely these exceptional values do give roots of unity. Therefore the powers of α are distinct whenever $r \notin \{0, 1, -1\}$: an equality $\alpha^j = \alpha^k$, with $j < k$, would give $\alpha^{k-j} = 1$. In particular the powers in (4) are distinct. For that single value one can also use its trace $\alpha + \bar{\alpha} = 6/5$, which cannot be an integer.

The nonzero rational exceptions $r = \pm 1$ will be handled separately using $\arctan(1) = \pi/4$, $\arctan(-1) = -\pi/4$, and the certified OpenAI bound for π .

3. Weighted logarithmic interpolation at distinct exponential coordinates

We first describe the geometric input, because it is the point at which the extension changes the original centers. The geometry and interpolation are over $\mathbb{C}$; the matrices used later have Gaussian rational entries.

Let $m, K \geq 1$, and let

$$W = (w_0, \dots, w_m), \quad V = (v_0, \dots, v_m)$$

be positive rational weights. Let $\sigma > 0$ be rational. Define the conservative comparison constant

$$\mathcal{C}(m, \sigma) = 2(m+2)^{m+2} \left(1 + \frac{m+2}{\sigma}\right)^{m+2}.$$

We impose the following conditions:

$$K(1 + 3\sigma)^{m+1} \frac{\prod_i w_i}{\prod_i v_i} < 1, \quad (1 + \sigma)w_i < v_i \quad (1 \leq i \leq m). \quad (\text{G1})$$

For equally sized subsets $A, B \subseteq \{0, \dots, m\}$, if the largest index in their symmetric difference is positive and belongs to A , require

$$\mathcal{C}(m, \sigma) \prod_{j \in B} v_j < \prod_{i \in A} w_i. \quad (\text{G2})$$

The centers are

$$b_j = (y_j, c_{j1}, \dots, c_{jm}), \quad 0 \leq j < K,$$

with distinct nonzero y_j . There is no requirement that the c_{ji} be distinct. Near each center use formal coordinates

$$t = Y/y_j - 1, \quad u_i = X_i - c_{ji} - \log(1 + t). \quad (\text{G3})$$

A jet packet of height H consists of the coefficients of $t^s u^\beta$ with $v_0 s + \sum_i v_i \beta_i < H$.

Interpolation theorem used here. There is a positive integer R , clearing the weight denominators, such that for every sufficiently large integer n , polynomials of W -weighted degree at most $H = nR$ map surjectively onto the packets of height H at all K centers.

The cofinal heights nR suffice for the determinant argument. We do not need, and do not claim as a checked intermediate theorem, surjectivity for every sufficiently large real height.

3.1 A curve estimate

For an irreducible algebraic curve C in the affine space with $Y \neq 0$, work on its smooth complete normalization. At every place P , set

$$D_W(C) = \sum_P \max \left(0, \frac{\text{pole}_P Y}{w_0}, \frac{\text{pole}_P X_1}{w_1}, \dots, \frac{\text{pole}_P X_m}{w_m} \right).$$

The notation means the sum of the displayed local maxima; identically zero coordinates contribute no poles. For each branch over a center define

$$h_P = \min \left(\frac{\text{ord}_P t}{v_0}, \frac{\text{ord}_P u_1}{v_1}, \dots, \frac{\text{ord}_P u_m}{v_m} \right), \quad S(C) = \sum_{P \text{ over centers}} h_P.$$

Orders of identically zero series may be infinite. On a nonconstant curve the minimum remains finite.

The needed estimate is

$$D_W(C) \geq (1 + \sigma)S(C). \quad (\text{G4})$$

Suppose instead that (G4) fails. The volume inequality in (G1) implies, by counting monomials and jet conditions, that for every sufficiently large N there is a nonzero polynomial F_N of W -degree at most N , whose V -weighted order at every center is at least $(1 + 3\sigma)N$. This count needs no independence of the vanishing conditions.

Use the commuting polynomial differential frame

$$D_0 = Y \partial_Y + \sum_{i=1}^m \partial_{X_i}, \quad D_i = \partial_{X_i} \quad (i > 0).$$

In the chart (G3), these become $(1+t)\partial_t, \partial_{u_i}$. Differentiation preserves the W -degree bound and loses at most its assigned V -cost in weighted order.

Every derivative word of cost at most σN therefore has weighted order at least $(1+2\sigma)N$. Its restriction to each branch has order at least $(1+2\sigma)Nh_P$, whereas its total pole degree is at most $ND_W(C)$. The assumed failure of (G4) forces all these derivatives to vanish identically on C .

Consider common zero loci of derivatives at successive cost cutoffs

$$r\delta_N, \quad \delta_N = \sigma N/(m+2).$$

Choose nested irreducible components containing the affine part of C . Their dimensions range from 1 to m ; consecutive stages eventually share a component Z . On this component, the defining equations from the earlier stage have degree at most N , and their further derivatives through cost δ_N vanish.

We now use the inherited **persistent normal comparison lemma**. For a coordinate normal basis indexed by A , and a differential-frame normal basis indexed by B , persistence gives

$$\prod_{i \in A} w_i \leq \mathcal{C}(m, \sigma) \prod_{j \in B} v_j. \quad (\text{G5})$$

The lemma compares an upper bound from weighted Bézout with a lower bound from rectangular transverse jets. Its large- N requirement is uniform in the component: the relevant rectangular cutoffs are at least 2 once N is sufficiently large in terms of the fixed weights, dimension and σ .

This is a substantial reused lemma, not a new result about arbitrary analytic frames. The formal applications are mapped in §9 to the original persistent-weight and transverse-slice sources.

If some positive-index coordinate direction is not tangent to Z , take the largest such index i . The coordinate normal basis can include i , with all larger directions tangent. Equations (G2) and (G5) force every frame normal basis to include D_i . All other frame vectors lie in

$$\ker(dX_i - dY/Y).$$

Their normal rank is one less than the codimension, so the tangent space of Z lies in this hyperplane. Thus $dX_i = dY/Y$ on Z .

If Y were nonconstant, restrict to an algebraic curve on which it remains nonconstant and pass to a smooth complete model. An exact differential dX_i has zero residues, while dY/Y has residue $\text{ord}_P Y$ at each zero or pole. A nonconstant rational function has such a place, giving a contradiction. Consequently Y is constant on Z .

In the remaining case all positive-index coordinate directions are tangent. Since Z is proper, its tangent space is their span; therefore $dY = 0$, and characteristic zero again makes Y constant. In either case Y is constant on C .

Distinct y_j now matter: a constant- Y curve meets at most one center. Some X_i is nonconstant on C . At that center $t = 0$ and $u_i = X_i - c_{ji}$, so the zero/pole equality for this function gives

$$v_i S(C) \leq w_i D_W(C).$$

The coordinate inequality in (G1) contradicts the assumed failure of (G4). This proves the curve estimate, counting every branch, including branches over singular points.

3.2 From curves to bounded-degree interpolation

Choose R so that $R/w_i, R/v_i$ are positive integers. Compactify by the monomials of W -degree at most R , retaining the constant coordinate as the affine chart. If $L = \mathcal{O}(1)$, then on every normalized affine curve,

$$\deg L = RD_W(C).$$

The constant and pure-coordinate monomials attain the local maxima defining this degree.

Replace the logarithm in (G3) by a polynomial truncation whose first omitted term has t -weight larger than every relevant v_i . This leaves the weighted filtration and branch contacts unchanged. At center j , form the finite-colength ideal

$$I_j = (t^{R/v_0}, (u'_1)^{R/v_1}, \dots, (u'_m)^{R/v_m}).$$

These ideals give a coherent ideal I , equal to the unit ideal away from the centers.

On the ordinary blowup, write $I\mathcal{O} = \mathcal{O}(-E)$ and $A = \pi^*L$. The source constructs the actual curve models and proves, for noncontracted affine curves,

$$A \cdot C = RD_W(C), \quad E \cdot C = RS(C).$$

Affine curves with $Y \equiv 0$ meet none of the prescribed centers, so $S(C) = 0$ and their curve estimate is automatic. Together with (G4) for curves meeting $Y \neq 0$, this makes $A - (1 + \sigma)E$ nef. Boundary curves avoid E ; on contracted curves A has degree zero and $-E$ is positive by relative ampleness.

For a sufficiently large integer $r > 1$, $rA - E$ is ample. The exact identity

$$A - E = \frac{r - 1}{r(1 + \sigma) - 1} [A - (1 + \sigma)E] + \frac{\sigma}{r(1 + \sigma) - 1} [rA - E] \quad (\text{G6})$$

has positive coefficients, and supplies ampleness together with a uniform positive curve margin. This step does not infer ampleness from strict nefness alone.

For all sufficiently large n , the ordinary blowup has

$$\pi_*\mathcal{O}(-nE) = I^n, \quad R^q\pi_*\mathcal{O}(-nE) = 0 \quad (q > 0).$$

These are eventual identities for ordinary powers of I ; they are not assertions about integral closures or about every small n . Serre vanishing for $A - E$, followed by the projection formula and Leray, yields

$$H^1(X, I^n L^n) = 0.$$

Sections therefore surject onto the local quotients. The monomials generating I_j^n have weighted order at least nR , so this surjection prescribes all packets below $H = nR$. Completion does not change these finite-colength quotients.

Finally, eventual polynomial generation for the projective embedding expresses sections of L^n as degree- n homogeneous polynomials in its coordinates. On the affine chart their W -degree is at most nR . This last step is essential: a Chinese remainder theorem without a degree bound would not give our interpolation statement.

4. The actual matrix

Fix the data in (2) and an exponent $\nu > 2$. Assume for contradiction that there are approximations p/q with arbitrarily large positive integer denominators and

$$|\xi - p/q| \leq q^{-\nu}.$$

Choose finitely many of them, denoted $r_i = p_i/q_i$, whose denominators will be taken very large and successively separated in §7. Set

$$w_i = \lceil \log q_i \rceil, \quad v_i = w_i/\theta \quad (1 \leq i \leq m),$$

with rational $0 < \theta < 1$. Use the multiplicative centers $y_j = \alpha^j$ and additive centers $c_{ji} = ijr_i$, for $0 \leq j < K$. Here the isolated symbol i before jr_i denotes the imaginary unit; subscripts index the chosen approximants. Equivalently, the additive center in coordinate k is $\sqrt{-1}jp_k/q_k$.

Choose $F_0 > 2/\theta$, and define

$$T_i = \lceil F_0 w_i / v_0 \rceil, \quad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1} t^k}{k}.$$

The omitted tail starts at order T_i , whose weight satisfies $T_i v_0 \geq F_0 w_i > v_i$. It therefore preserves the jet packets in §3.

Columns are monomials $Y^h X^\gamma$ with $w_0 h + w\gamma \leq H$. Rows are triples (j, s, β) with

$$0 \leq j < K, \quad v_0 s + w\beta/\theta < H.$$

The entry in the actual matrix is

$$[t^s u^\beta] (\alpha^j (1+t))^h \prod_i (\sqrt{-1} jr_i + G_i(t) + u_i)^{\gamma_i}. \quad (5)$$

The exponential factor α^{jh} is retained. All entries lie in $\mathbb{Q}(i)$.

The interpolation theorem supplies full row rank at cofinal heights $H = nR$. Select a square nonzero minor Δ_H containing every row. Let M_H be the number of rows, and write

$$b_H = \frac{\sum_{\text{rows}} w\beta}{M_H H}, \quad 0 \leq b_H \leq \theta.$$

Weighted lattice counting gives

$$M_H \sim \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}. \quad (6)$$

Both determinant bounds below concern this same minor, and are uniform in the selected columns.

5. Gaussian arithmetic and the row savings

Let

$$L_i = \text{lcm}(1, \dots, T_i), \quad E_H = \prod_i L_i^{\lfloor H/w_i \rfloor}.$$

Multiply column (h, γ) by $D^{Kh} \prod_i q_i^{\gamma_i}$, row (j, s, β) by $\prod_i q_i^{-\beta_i}$, and every entry by E_H . The row factors are essential; discarding them loses the leading saving in (8).

When $\beta \leq \gamma$, the cleared entry is

$$E_H U^{jh} D^{(K-j)h} \binom{\gamma}{\beta} [t^s] (1+t)^h \prod_i (\sqrt{-1} j p_i + q_i G_i(t))^{\gamma_i - \beta_i}. \quad (7)$$

Otherwise it is zero. The identity $D^{Kh} \alpha^{jh} = U^{jh} D^{(K-j)h}$ uses $j \leq K$. Each $\gamma_i - \beta_i \leq \lfloor H/w_i \rfloor$, so the displayed lcm powers clear all coefficient denominators. Thus the cleared entries, and their determinant, lie in $\mathbb{Z}[i]$.

All scaling factors are nonzero. Since $\Delta_H \neq 0$, its cleared determinant is a nonzero Gaussian integer Z . Consequently $|Z|^2$ is a positive integer and $|Z| \geq 1$. This does not introduce a degree-two multiplier: the two complex embeddings give the same absolute value, and the lower bound is already in the chosen complex absolute value.

Put $\Lambda = 4 + \log 4$ and $w_* = \min_i w_i$. The inherited conservative lcm estimate is $\log L_i \leq \Lambda T_i$. Column degree, $T_i \leq F_0 w_i / v_0 + 1$, and $w_i - 1 \leq \log q_i \leq w_i$ give

$$\frac{\log |\Delta_H|}{M_H H} \geq -(1 - b_H) - E_{\text{ar}}, \quad (8)$$

where

$$E_{\text{ar}} = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*} + \frac{K \log D}{w_0}. \quad (9)$$

Indeed, the common clearing factor costs at most the first two terms. The selected columns cost at most $M_H H (1 + K \log D / w_0)$. The row logarithms save at least $M_H H (b_H - \theta / w_*)$. Combining these three inequalities yields (8). The Gaussian numerator needs no further clearing factor and contributes no additional lower-bound cost.

6. Complex analytic expansion and collision saving

Set $\omega = i\xi$ and $\rho = 100 \max(1, |\xi|)$. For a column monomial $P = Y^h X^\gamma$ and multi-index b , define the entire function

$$f_{b,P}(z) = [u^b] P(e^z, z + u_1, \dots, z + u_m) = \binom{\gamma}{b} e^{hz} z^{|\gamma| - |b|},$$

with value zero if $b \not\leq \gamma$. On $|t| < 1$, put

$$z_j(t) = j\omega + \log(1+t).$$

Since $e^{j\omega} = \alpha^j$, we have $e^{z_j(t)} = \alpha^j (1+t)$. The additive arguments in (5) become

$$z_j(t) + u_i + e_{ji} + \tau_i(t), \quad e_{ji} = \sqrt{-1} j(r_i - \xi), \quad \tau_i = G_i - \log(1+t). \quad (10)$$

No step replaces $j\omega$ by the principal value of $\log(\alpha^j)$. Such a replacement would be invalid when the powers wind around the unit circle; it is absent from both the argument and the formalization.

Expanding (10) gives a finite linear combination of row vectors

$$([t^\ell] f_{b,P}(z_j(t)))_P,$$

with $b \geq \beta$, $wb \leq H$, and coefficients independent of the selected column. The rotation has modulus one, so the approximation error satisfies

$$|e_{ji}| = j|r_i - \xi| \leq 2Ke^\nu e^{-\nu w_i}.$$

Each tail starts at T_i . A term using tail multiplicities e_i can contribute only if $\sum_i e_i T_i \leq s$, whence $\sum_i w_i e_i < H/F_0$. On the circle $|t| = 1/2$, the inherited coefficient and binomial estimates bound its scalar by

$$\exp\{-\nu w(b - \beta) + HE_{\text{tr}}\}, \quad E_{\text{tr}} = \frac{\nu}{F_0} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K) + \nu}{w_*}. \quad (11)$$

This accounts for the approximate additive centers; replacing them by exact exponential centers would omit this term.

On $|z| \leq \rho K$, the entire functions are bounded by

$$\exp\left\{H\left(\frac{\rho K}{w_0} + \frac{\log(2\rho K)}{w_*}\right)\right\}.$$

The center estimate $|j\omega| + 3/4 \leq \rho K/2$ places $z_j(t)$ within the smaller disc when $|t| = 1/2$. Taylor expansion in $z/(\rho K)$ is therefore absolutely convergent with a geometric majorant. If two rows use the same transverse index b and the same Taylor order, that determinant term is zero. Among n_b rows with a common b , the Taylor orders of a surviving term have sum at least $\binom{n_b}{2}$.

The inherited collision determinant estimate, with conservative constant $c = \log 2/4$, gives

$$|\det B| \leq \exp\left\{-c \sum_b n_b^2 + M_H H(E_{\text{hol}} + o(1))\right\},$$

where

$$E_{\text{hol}} = \frac{\rho K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2\rho K)}{w_*}, \quad E_{\text{an}} = E_{\text{tr}} + E_{\text{hol}}. \quad (12)$$

The estimate also holds for singular test matrices. Its proof expands the determinant of the entire-function row tests, uses the repeated-order cancellation just described, and bounds the remaining absolutely convergent geometric sums and factorial factors. The formal source applies the existing theorem for arbitrary complex ω , retaining the factor $\exp(j\omega)^h = \alpha^{jh}$.

Choose $A > \theta$ and $0 < \eta < 1$. Let $N_A(H)$ count multi-indices with $wb \leq AH$, and set

$$c_H = \frac{c\eta^2 M_H}{HN_A(H)}.$$

If at least ηM_H rows use those low indices, Cauchy–Schwarz gives $\sum_b n_b^2 \geq \eta^2 M_H^2 / N_A(H)$, hence the normalized saving $-c_H$. Otherwise more than $(1 - \eta)M_H$ rows have $wb > AH$, and the product of scalar factors in (11) gives the saving $-\nu(A(1 - \eta) - b_H)$.

For fixed parameters, the finite expansion has at most

$$Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)$$

choices per row. Determinant multilinearity introduces at most $Q_H^{M_H}$ choices, adding only $\log Q_H/H = o(1)$ after normalization. Together with the normalized factorial remainder, this proves

$$\frac{\log |\Delta_H|}{M_H H} \leq E_{\text{an}} + o(1) + \max\{-c_H, -\nu(A(1 - \eta) - b_H)\}. \quad (13)$$

All parameters and approximations are fixed before H tends to infinity. Lattice counting also gives

$$c_H \longrightarrow c_\infty = \frac{c\eta^2 K \theta^m}{(m+1)v_0 A^m}. \quad (14)$$

7. Parameter choice and contradiction

Fix ξ, U, D, ν first. The quantities ρ and $\delta = \log D$ are fixed. Choose positive rational θ, A, B, C such that

$$0 < \theta < A < B < 1, \quad \nu(A - \theta) > 1 - \theta, \quad C > 1, \quad CB < 1, \quad B < C\theta < 1. \quad (15)$$

These inequalities are compatible precisely with the required range $\nu > 2$ in this construction: choose rational $b \in (1/2, 1 - 1/\nu)$ and small rational $s > 0$, then set $\theta = 1 - s$, $A = 1 - bs$. The approximation gap is positive and $A^2 < \theta$ for sufficiently small s . Take $C \in (A/\theta, 1/A)$, then $B \in (A, \min(1/C, C\theta))$.

Take small rational $\eta > 0$ with

$$g = \nu(A(1 - \eta) - \theta) - (1 - \theta) > 0, \quad \epsilon = \min(g/2, 1/2).$$

Choose $F_0 > 2/\theta$ with $\nu/F_0 < \epsilon/3$. Now increase m , setting

$$K = \lfloor C^m \rfloor, \quad w_0 = B^{-m}, \quad v_0 = 2K\theta^m w_0.$$

The ambient interpolation mass is $1/2$, while

$$K/w_0 \leq (CB)^m \rightarrow 0, \quad v_0 \sim 2(C\theta/B)^m \rightarrow \infty, \quad m/v_0 \rightarrow 0,$$

and

$$c_\infty = \frac{c\eta^2 (B/A)^m}{2(m+1)} \rightarrow \infty.$$

Choose one finite $m \geq 1$ for which

$$\frac{\Lambda F_0 m + 2 \log 2}{v_0} + \frac{(\rho + \delta)K}{w_0} < \epsilon/3, \quad c_\infty > 2. \quad (16)$$

Then fix these parameters. Since $K\theta^m < 1$, choose a small positive rational σ with

$$(1 + 3\sigma)^{m+1}/2 < 1, \quad (1 + 3\sigma)^m K\theta^m < 1, \quad (1 + \sigma)\theta < 1.$$

Only now choose the finitely many approximations. Arbitrarily large denominators permit all weights to be large and successively separated, meeting the product comparisons in §3 and

$$\Lambda \sum_i \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(2\rho K)}{w_*} < \epsilon/3. \quad (17)$$

There are finitely many product comparisons. Common factors cancel because $v_i = w_i/\theta$ for positive indices, and increasing the next denominator meets each remaining comparison. The interpolation threshold may depend on the chosen centers; that causes no circularity, since the centers are fixed before the final limit in H .

Equations (9), (11), (12), (16), and (17) imply

$$E_{\text{ar}} + E_{\text{an}} < \epsilon < g, \quad c_{\infty} > 1 + E_{\text{ar}} + E_{\text{an}}. \quad (18)$$

In particular the denominator cost $K \log D/w_0$ and complex radius cost $\rho K/w_0$ both tend to zero at the dimension-selection stage. There is no unaccounted conjugate term.

For sufficiently large cofinal $H = nR$, choose the nonzero minor furnished by interpolation. If the maximum in (13) is the collision alternative, compatibility with (8) requires

$$c_H \leq 1 - b_H + E_{\text{ar}} + E_{\text{an}} + o(1) \leq 1 + E_{\text{ar}} + E_{\text{an}} + o(1),$$

contradicting (14) and (18). In the other alternative, compatibility requires

$$\nu(A(1 - \eta) - b_H) - (1 - b_H) \leq E_{\text{ar}} + E_{\text{an}} + o(1).$$

The left side is at least g , since it decreases with b_H and $b_H \leq \theta$. This contradicts (18) as well. Thus arbitrarily large approximations of the assumed quality cannot exist, proving the angle statement (2).

8. Scaling, the exceptional cases, and the exponent

We use an explicit transport of the approximation bound. Suppose x satisfies (1) and n is a positive integer. Fix $\nu > 2$, and choose $\mu = (\nu + 2)/2$, so $2 < \mu < \nu$. For sufficiently large q , the bound for x and $q^{\nu-\mu} \geq n$ give

$$\left| \frac{x}{n} - \frac{p}{q} \right| = \frac{1}{n} \left| x - \frac{np}{q} \right| \geq \frac{1}{n} q^{-\mu} \geq q^{-\nu}. \quad (19)$$

Negation preserves the same statement by replacing p with $-p$. These are formal lemmas on the fully quantified eventual bound; no invariance under algebraic scaling is presumed.

For $r \notin \{0, 1, -1\}$, §2 gives the data (2) for $x = 2 \arctan r$, and (19) with $n = 2$ proves (1). The dedicated $r = 1/2$ path uses (4). For $r = \pm 1$, apply (19) with $x = \pi$, $n = 4$, using OpenAI's certified π bound, and use negation for -1 . This proves (1) for every rational $r \neq 0$.

Finally, if $\arctan r = u/v$ were rational, the unreduced fractions $ku/(kv)$ would violate (1) at arbitrarily large denominators. Hence it is irrational. Dirichlet approximation supplies infinitely many reduced approximants of exponent 2, while (1) excludes every larger exponent. Therefore its irrationality exponent is exactly 2.

9. Formal boundary, verification, and attribution

The endpoint source is [Arctangent/Main.lean](#). The internal generic angle theorem retains the data in (2), but both public arctangent endpoints discharge them. The family endpoint has only the hypothesis $r \neq 0$.

- Specified angle and Gaussian rational base: [PeriodData](#)
- Rational-coordinate roots of unity: [GaussianRoots](#)
- Double-angle exponential, all rational inputs: [ArctanCenters](#)
- Distinct-center interpolation geometry: [Inherited Interpolation](#), [GlobalMatrixInterpolation](#)
- Gaussian clearing and exact row savings: [ScaledArithmetic](#)

- Rotated approximation errors: [ScaledApproximationErrors](#)
- Complex collision estimate: [Inherited ScaledCollision](#), [LiteralAnalyticSummand](#)
- Full determinant expansion and upper bound: [AnalyticAggregate](#)
- All parameter costs and final contradiction: [Inherited AdmissibleParameters](#), [Parameters](#), [DeterminantContradiction](#)
- Division and negation transport: [Scaling](#)
- Inherited normal comparison: [PersistentWeightComparison](#), [TransverseSliceComparison](#)

The new Lean sources are separate from the preceding audited rational-logarithm bundle. The source manifest, compiler and checker pins, reproduction commands, hashes, verification results, and negative controls are summarized in [VERIFICATION.md](#). Those records distinguish the initial Lean build, transitive axiom audits, independent exported-proof checking, and replay through a second stock Lean kernel. They do not claim that a human referee has reconstructed the proof or that the priority search is exhaustive.

The mathematical framework and original library are credited to [OpenAI/math](#), [commit adc7f1241b42e322a6451854ab7e4b4c146bf78a](#). Our preceding positive-rational-logarithm package supplies the distinct-center geometric generalization and generic complex analytic interfaces used here. We make no exhaustive priority or novelty claim.

AI use disclosure. Ryan Matthew Casper initiated and directed the investigation. OpenAI coding agents contributed substantially to assessing the Gaussian extension, developing the mathematical argument, adapting and writing the Lean proofs, running verification and automated checks, and drafting this exposition. The work reuses OpenAI’s method and released library, Mathlib, and the author’s preceding AI-assisted rational-logarithm formalization. Automated review and kernel-checking results are evidence about the specified formal statements; they are not external human peer review. No external referee was contacted in preparing this package.

The central new checks are the exact Gaussian clearing identity (7), preservation of the row savings, the imaginary approximation-center error (10), the retained exponential factor in the analytic matrix, all fixed-base costs in (16), and the explicit treatment of torsion and scaling. The inherited geometric comparison and ordinary-power interpolation remain substantial parts of the argument and should be included in any mathematical review.